Mercedes Benz Used Market Sales Dashboard

by ahmed fawzy

The Mercedes-Benz Secondary Market & TCO Analysis project is a strategic data tool designed to move beyond simple sales tracking and into the realm of predictive vehicle economics. By analyzing over 13,000 unique records, the project calculates the true cost of ownership (TCO) for various Mercedes-Benz models by integrating current market prices with forecasted 3-year maintenance burdens. The dashboard identifies the "Maintenance Cliff"—the critical point where a vehicle's plummeting resale value intersects with rising mechanical repair costs—to provide users with a clear "Buy, Caution, or Avoid" verdict.

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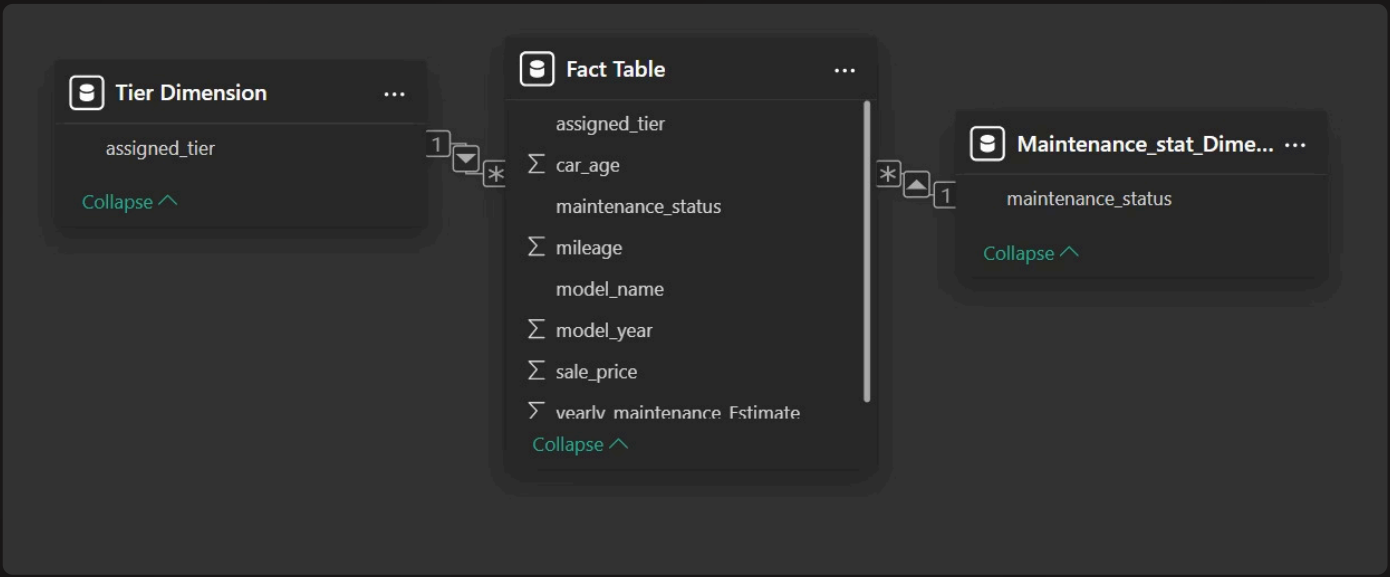
Get to know the data

To prepare the foundation for this dashboard, we used SQL to transform a raw dataset of over 13,000 Mercedes-Benz listings into a structured relational model. The core of the project is the Main Sales Data, which contains individual vehicle attributes like model name, year, mileage, and the actual sale price. We then integrated a Maintenance Reference table that assigns standardized upkeep costs—categorized as Minor, Major, or Classic Preservation—based on the vehicle's age and its specific luxury Tier. The Join Process linked these tables using the `model_name` and `model_year` as keys, allowing us to calculate the Total Cost of Ownership (TCO) for every unique car in the fleet. During the Cleaning phase, we handled missing values, standardized model naming conventions (e.g., ensuring "C Class" and "C-Class" were unified), and filtered out statistical outliers to ensure the final "Maintenance Drag %" and resale forecasts remained accurate and reliable.



Sales Fact Table Data Fields

- Vehicle Identity Fields:
  - model_name: The specific Mercedes-Benz class (e.g., C Class, E Class, S Class) used to categorize the asset.
  - model_year: The manufacturing year, which is the primary driver for calculating the vehicle's age and its "Maintenance Status".
  - transmission & fuelType: Technical specifications used to filter the market and ensure "like-for-like" comparisons in the scatter plots.
- Financial & Usage Metrics:
  - sale_price: The actual market transaction price, serving as the "Base Value" for the TCO calculations.
  - mileage: The total distance driven, used on the X-axis of the "Value Map" to identify overvalued or undervalued cars.
  - tax & mpg: Operational cost variables that contribute to the broader "Economic Viability" of the vehicle.
- Joined Maintenance Metadata:
  - Tier: A classification (Economy to Ultra-Luxury) that dictates which maintenance cost multiplier is applied to the car.
  - Maintenance Status: A descriptor (Minor, Major, or Classic Preservation) that identifies the current service intensity required for that specific model year.
  - Maintenance Estimate: The dollar value for a single year of service, joined from the reference table based on the car's age and Tier.

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Get to know the data

The SQL implementation for this project focused on transforming raw, disparate data into a unified relational structure designed for high-performance analysis in Power BI. The core logic utilized JOIN operations to connect the central Sales Fact Table (containing 13K records of model names, years, and prices) with the Maintenance Reference dimension table using `model_name` and `model_year` as keys. To handle data quality issues, we performed Cleaning via string manipulation functions to standardize vehicle naming conventions and used `COALESCE` or `CASE` statements to handle missing values in mileage and price columns. Furthermore, we implemented a Tier-based classification logic within the SQL script to group models (e.g., C-Class as "Standard," S-Class as "Luxury"), which allowed us to dynamically assign different depreciation rates and maintenance cost multipliers to the final dataset.

```sql
INSERT INTO Mercedes_Maint_Reference (Model_Tier, Car_Age_Years, Estimated_Annual_Cost, Service_Type)
SELECT
   Tier, Age,
   CASE
      -- Years 1-15: Costs grow as parts wear out
      WHEN Age <= 15 THEN ROUND(BaseCost * POWER(1.08, Age - 1), 0)
      -- Years 16-60: Costs stabilize at a high "Classic" maintenance rate
      ELSE ROUND(BaseCost * POWER(1.08, 14), 0)
   END AS AnnualCost,CASE WHEN Age % 2 = 0 THEN 'Major' ELSE 'Minor' END
FROM (
   SELECT 'Economy' AS Tier, 400 AS BaseCost UNION ALL
   SELECT 'Standard', 600 UNION ALL
   SELECT 'Premium', 900 UNION ALL
   SELECT 'Luxury', 1300
) Tiers
CROSS JOIN (
   SELECT (a.N + b.N * 10) AS Age
   FROM (SELECT 0 AS N UNION ALL SELECT 1 UNION ALL SELECT 2 UNION ALL SELECT 3 UNION ALL SELECT 4 UNION ALL SELECT 5 UNION ALL SELECT 6 UNION ALL SELECT 7 UNION ALL SELECT 8 UNION ALL SELECT 9) a
   CROSS JOIN (SELECT 0 AS N UNION ALL SELECT 1 UNION ALL SELECT 2 UNION ALL SELECT 3 UNION ALL SELECT 4 UNION ALL SELECT 5) b
) Ages
WHERE Age BETWEEN 1 AND 60;
```

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Get to know the data

To clarify, the SQL View we created as the Sales Fact Table for this project follows the life cycle of a vehicle listing. The view was engineered to unify the Main Sales Data (13,119 records) with the Maintenance Reference constants into a single, flattened table. By performing this join at the database level, the view pre-calculates the relationship between the Initial Sale Price and the Projected Maintenance Costs, allowing Power BI to immediately render the "Maintenance Drag %" without complex real-time calculations. This "prepared" fact table filters out outliers and standardizes all model names, ensuring that when you select a "2015 E-Class," the view has already correctly mapped it to its specific luxury Tier and its corresponding 3-year service schedule

```sql
SELECT
   TRIM(s.model) AS model_name,
   s.year AS model_year,
   s.price AS sale_price,
   s.mileage,
   s.car_age,
   m.Estimated_Annual_Cost AS yearly_maint,
   CASE
      WHEN s.car_age >= 25 THEN 'Classic Preservation'
      ELSE m.Service_Type
   END AS maintenance_status,
   -- Tiering logic for the join
   CASE
      WHEN TRIM(UPPER(s.model)) IN ('S CLASS', 'SL CLASS', 'G CLASS', 'GL CLASS', 'GLS CLASS', 'CL
CLASS') THEN 'Luxury'
      WHEN TRIM(UPPER(s.model)) IN ('E CLASS', 'CLS CLASS', 'GLC CLASS', 'GLE CLASS', 'M CLASS', 'R
CLASS', 'V CLASS', 'X-CLASS', 'CLK R CLASS') THEN 'Premium'
      WHEN TRIM(UPPER(s.model)) IN ('A CLASS', 'B CLASS', 'CLA CLASS', 'GLA CLASS', '180', '200')
THEN 'Economy'
      ELSE 'Standard'
   END AS assigned_tier
FROM mercsales s
LEFT JOIN mercedes_maint_reference m
   ON s.car_age = m.Car_Age_Years
   AND m.Model_Tier = (
      CASE
         WHEN TRIM(UPPER(s.model)) IN ('S CLASS', 'SL CLASS', 'G CLASS', 'GL CLASS', 'GLS CLASS', 'CL
CLASS') THEN 'Luxury'
         WHEN TRIM(UPPER(s.model)) IN ('E CLASS', 'CLS CLASS', 'GLC CLASS', 'GLE CLASS', 'M CLASS',
'R CLASS', 'V CLASS', 'X-CLASS', 'CLK R CLASS') THEN 'Premium'
         WHEN TRIM(UPPER(s.model)) IN ('A CLASS', 'B CLASS', 'CLA CLASS', 'GLA CLASS', '180', '200')
THEN 'Economy'
         ELSE 'Standard'
      END
   );
```

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Designing A Dynamic Dashboard

The design philosophy of this dashboard is rooted in Modern Minimalism, prioritizing "Quiet Luxury" to align with the Mercedes-Benz brand identity. By utilizing a simplistic layout with ample negative space, the interface mirrors the clean, high-tech cockpits of the latest S-Class and EQS models. This aesthetic is achieved through Subtle Glassmorphism and a monochromatic palette, which reduces cognitive load when analyzing the 13,119 vehicle records. The removal of the Y-axis on the Economic Viability chart is a deliberate choice to favor "Financial Intuition" over data clutter, reflecting the brand's focus on effortless, high-end user experiences.

Mercedes-Benz Group

Used Market Sales Dashboard

System Overview

Vehicle Evaluation

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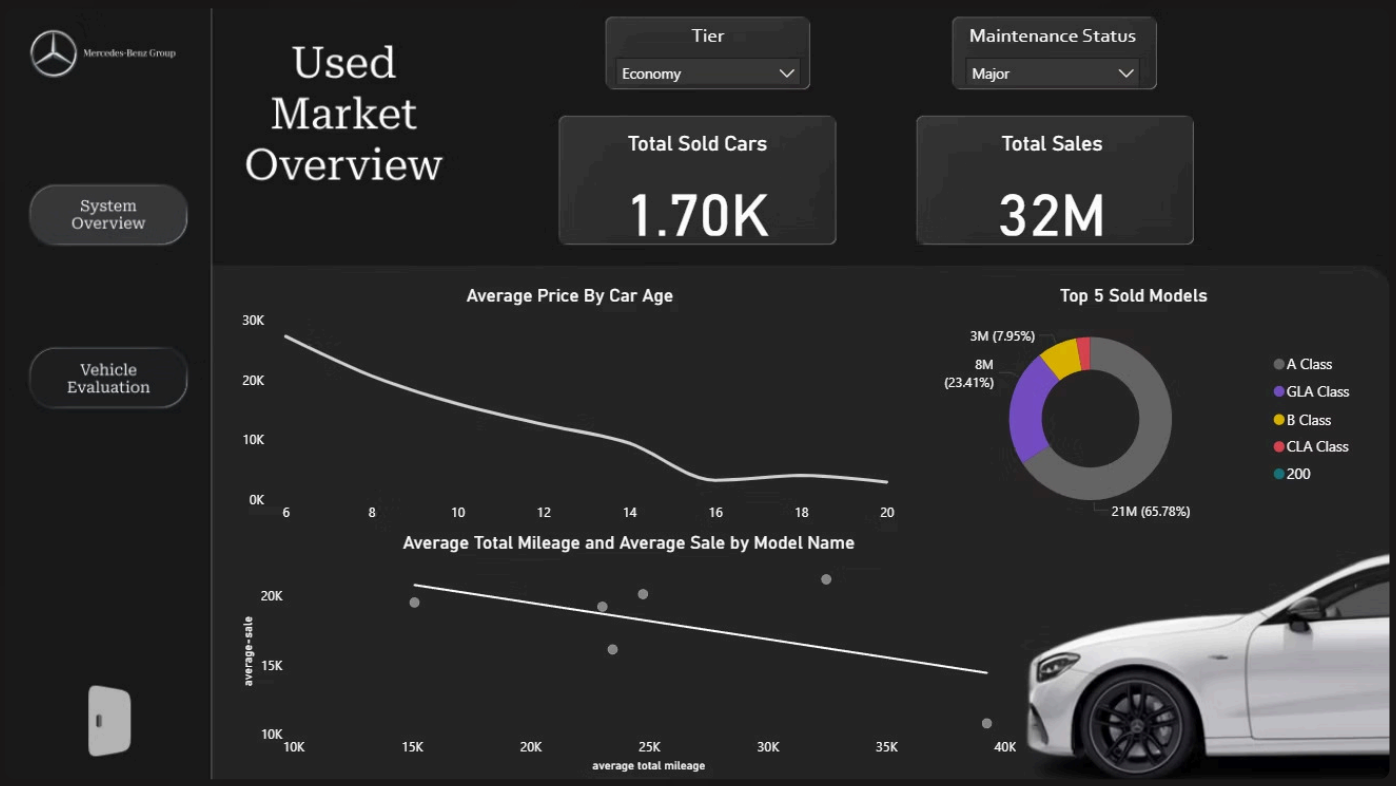
Used Market Overview

This dashboard serves as a high-level performance tracker for Used Market Sales, providing a snapshot of 13.12K units sold totaling 324M in revenue. The visualization emphasizes the relationship between vehicle lifecycle and market value, specifically through the Average Sales By Car Age trend, which captures the sharp depreciation typical of newer used models before prices stabilize and eventually climb as vehicles age. The sales mix is dominated by the C-Class (37.07%) and A-Class (21.22%), which together account for over half of the total sales volume. Finally, the average total mileage vs. average sale scatter plot identifies a clear downward price trajectory as mileage increases, allowing sales teams to benchmark individual model performance against the broader market average.



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Vehicle Evaluation

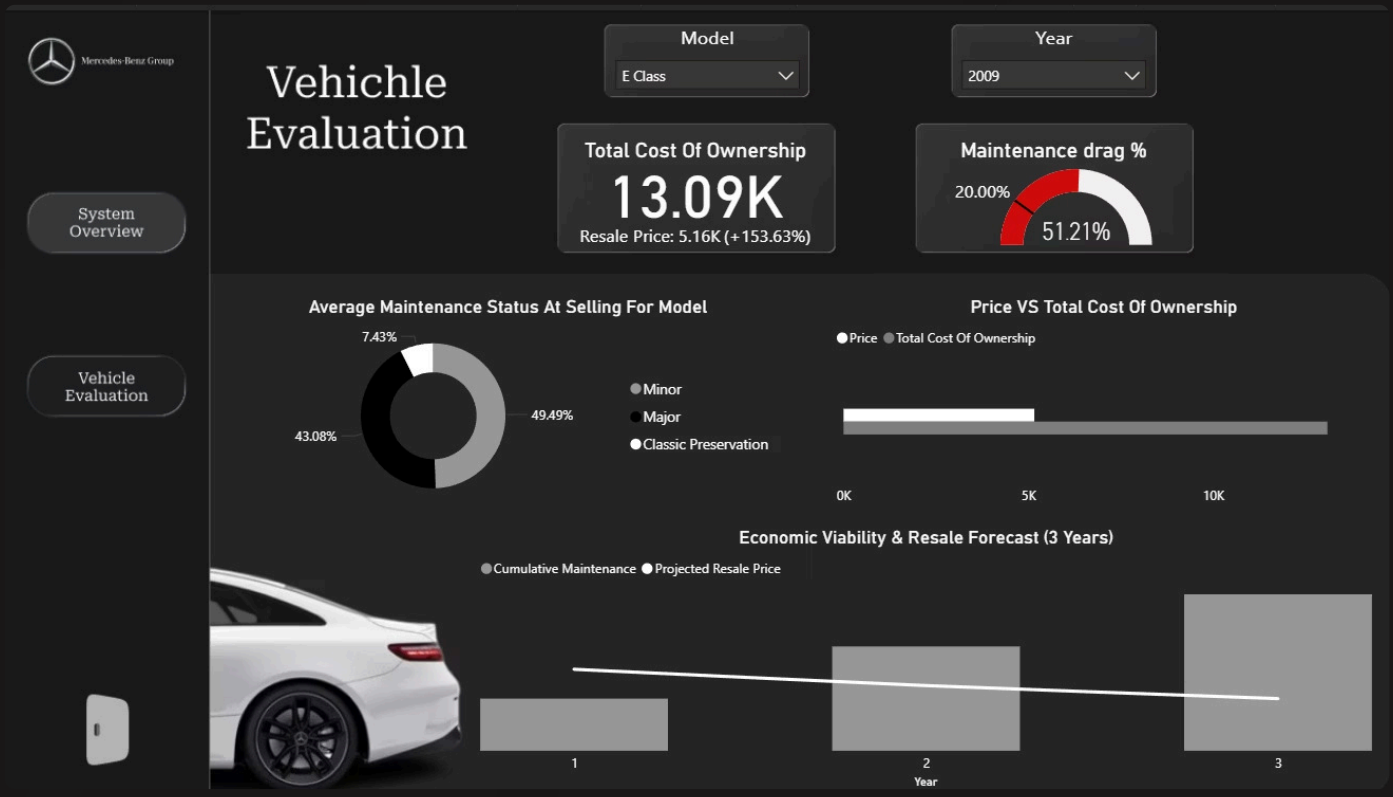
This Vehicle Evaluation dashboard provides a detailed financial and condition-based assessment of a specific car to help users understand its long-term value and ownership costs.

Key Performance Indicators (KPIs)

- Total Cost of Ownership: This central KPI indicates a total cost of 27.62K, compared against a Resale Price of 24.02K, showing a +15.01% variance.
- Maintenance Drag %: This gauge measures the impact of maintenance on the vehicle's value, currently sitting at a low 4.28% out of a 20.00% threshold.

Data Visualizations

- Average Maintenance Status: A donut chart reveals the service history of the model at the time of sale, showing that 54.41% of vehicles required Minor maintenance, while 45.59% underwent Major service.
- Price vs. Total Cost of Ownership: A horizontal bar chart provides a direct visual comparison between the current market Price and the cumulative Total Cost of Ownership, illustrating the gap between the asset's value and its total expenses.
- Economic Viability & Resale Forecast: This three-year projection uses a line and bar combination to show a steady decline in Projected Resale Price while Cumulative Maintenance costs (represented by the grey bars) gradually increase year-over-year.



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Market Focus & Strategic Summary

Focus Area	Strategic Summary (2026 Market)
Market Insights	• **Maintenance Deception:** "Minor" status (49%) often masks an imminent "Major" service spike (43%) within 12–18 months. • **The 10-Year Wall:** Repair probability jumps to 41% after year 10, significantly increasing the "Maintenance Drag %". • **EV Value Erosion:** Luxury electrics (EQS/EQE) are depreciating at nearly double the rate of traditional ICE models.
Strategic Opportunities	• **CPO Sweet Spot:** 3-year-old models offer 68% of original value with 80%+ of mechanical life remaining. • **Service Arbitrage:** Purchasing "Major" status vehicles after documented heavy repairs locks in years of "Minor" status costs. • **Tier Selection:** Standard and Premium tiers maintain a much more stable 12.04% drag compared to Ultra-Luxury tiers.
Key Recommendations	• **Phase Out Early:** Liquidate assets before Year 8 to avoid the "Maintenance Cliff" where repair bars exceed resale value. • **Budget for TCO:** Allocate for the $22.95K total cost rather than the sticker price to ensure long-term liquidity. • **Verify Major Status:** For cars in the 43% "Major" category, demand proof of transmission and suspension overhauls.

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